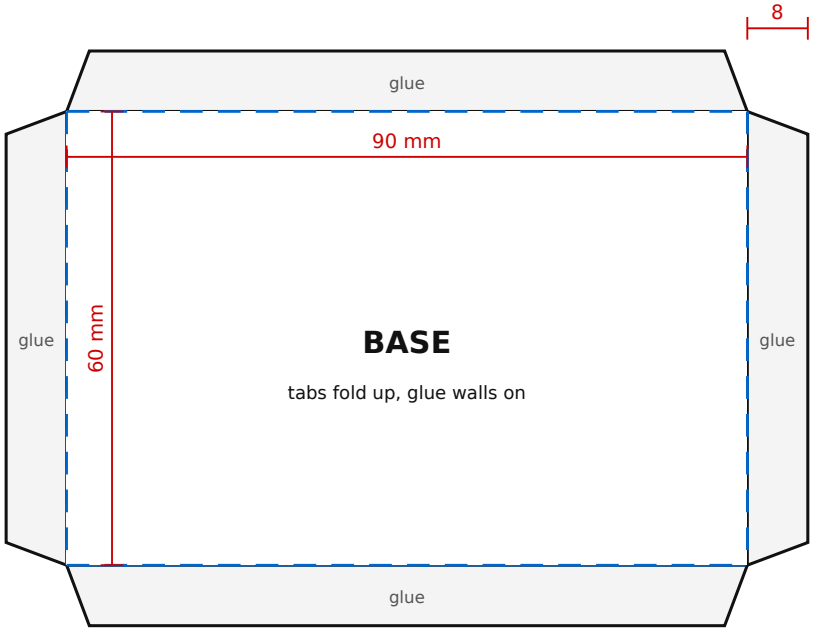
Robotics Project #13 · Templates 1/4 (1:1 scale)

Wheelbarrow Walker Robot

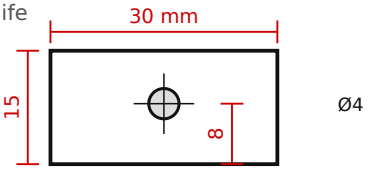
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

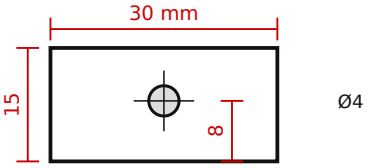
3 mm corrugated cardboard · grown-up cuts with craft knife



Barrow base 90x60 + 8 mm glue tabs

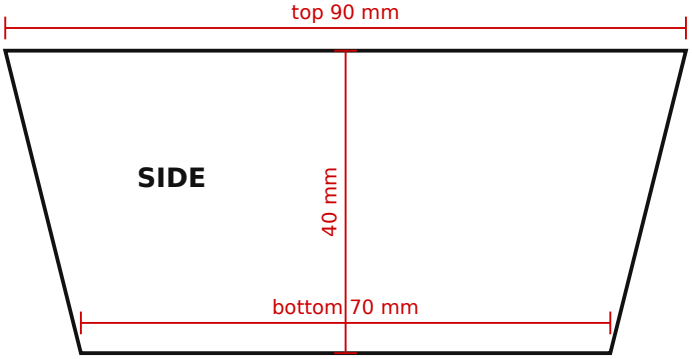


Axle hanger 1

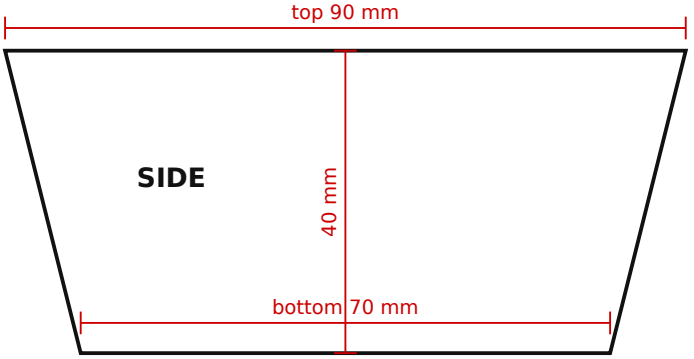


Axle hanger 2

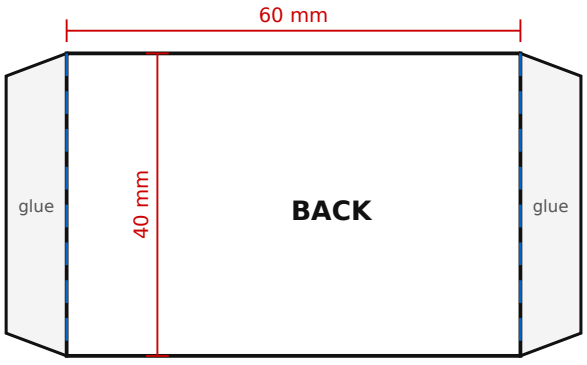
hole 8 mm up from bottom edge



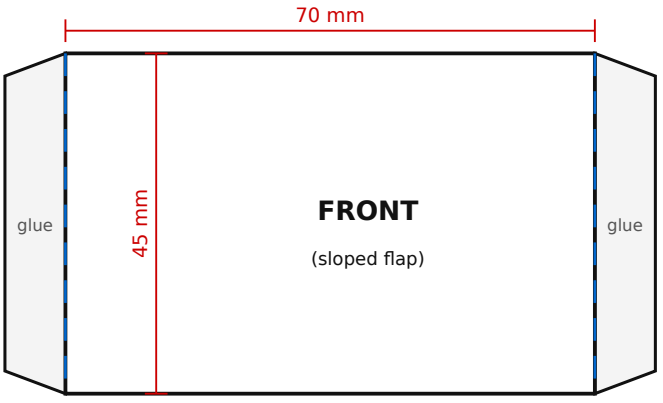
Side wall 1 (trapezoid)



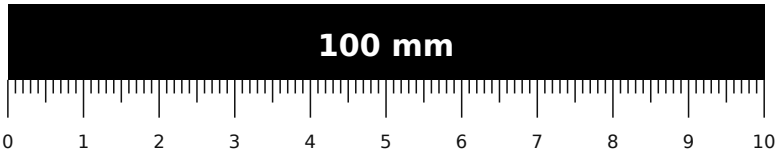
Side wall 2 (trapezoid)



Back wall 60x40 + tabs



Front flap 70x45 + tabs



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm.
If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.

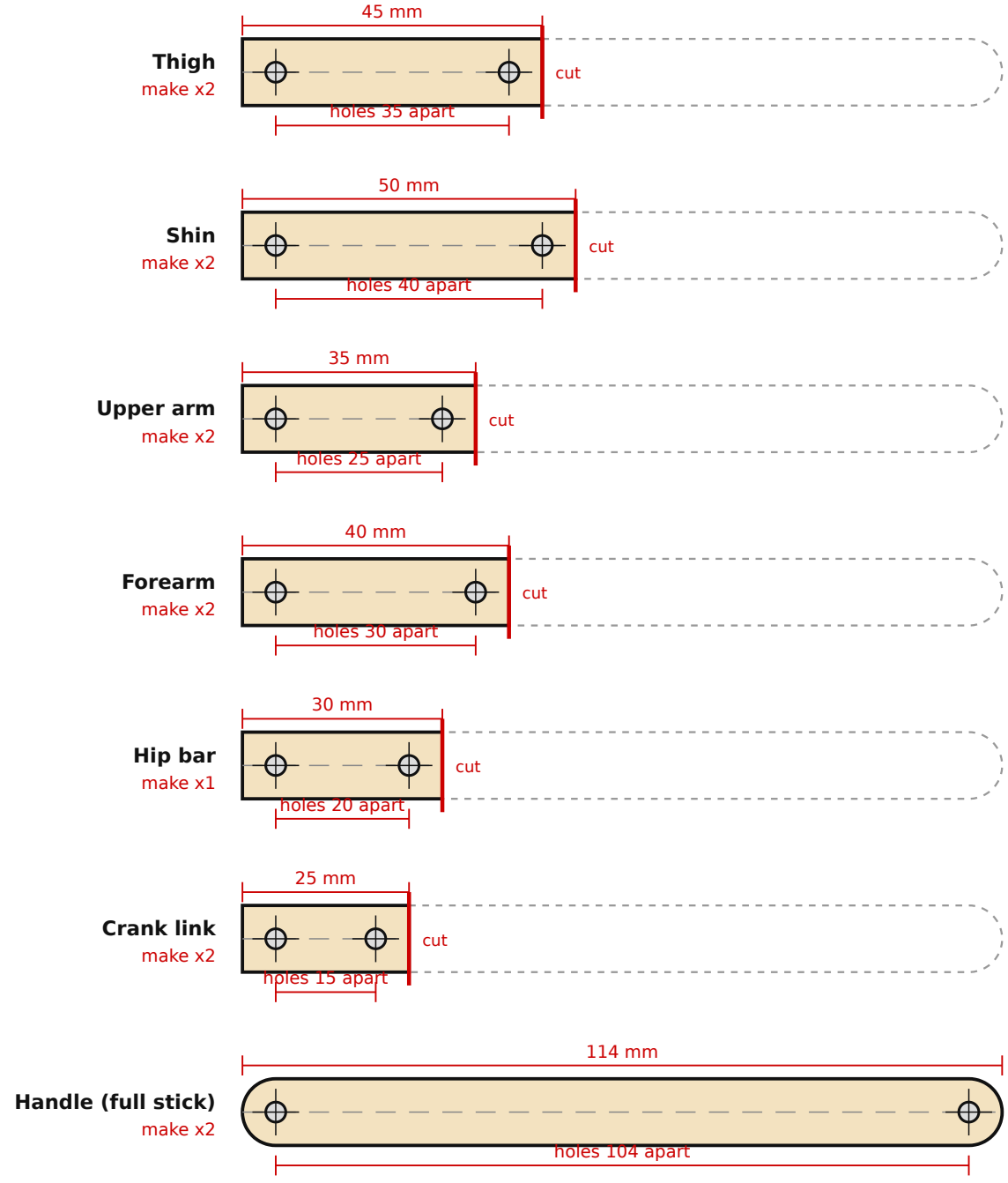
Robotics Project #13 · Templates 2/4 (1:1 scale)

Wheelbarrow Walker Robot

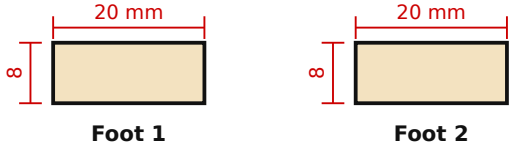
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

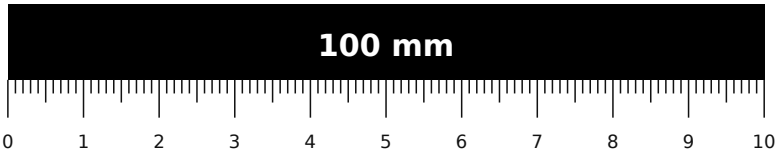
Lay an ice-cream stick (114x10) on each guide, mark the cut at the red line, grown-up cuts.



Ø3 pin holes 5 mm from each end (centre lines shown). Pin with toothpick / skewer bits so joints swing.



Feet: cut from stick offcuts



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm.
If not, reprint at 100% (Actual size).
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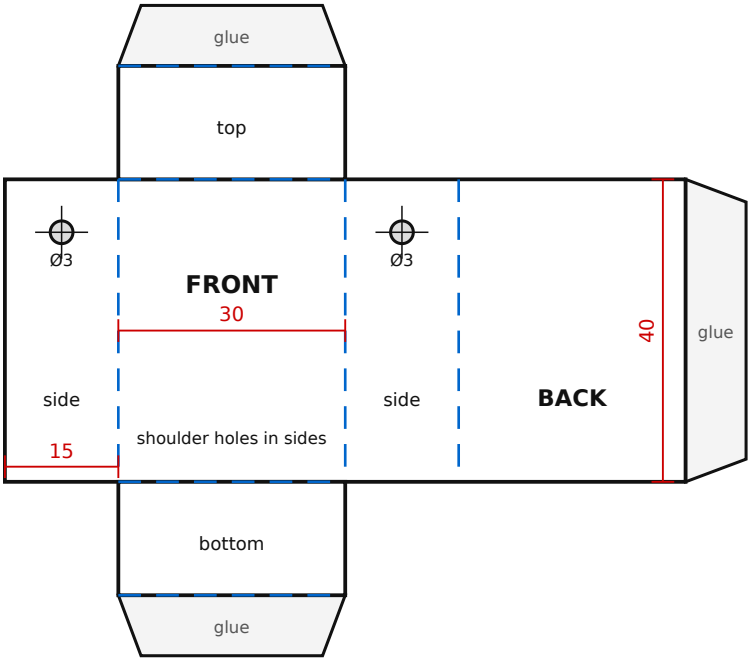
Robotics Project #13 · Templates 3/4 (1:1 scale)

Wheelbarrow Walker Robot

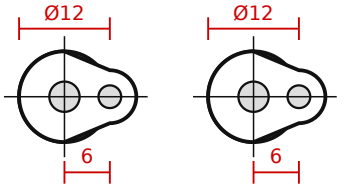
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

Card or corrugated cardboard (crank discs: 3 mm corrugated, glue 2 layers each)

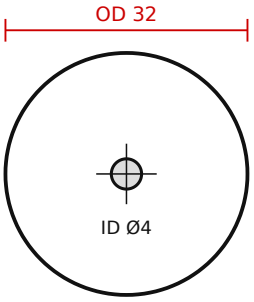


Torso box net 30x40x15 (fold, glue tabs)

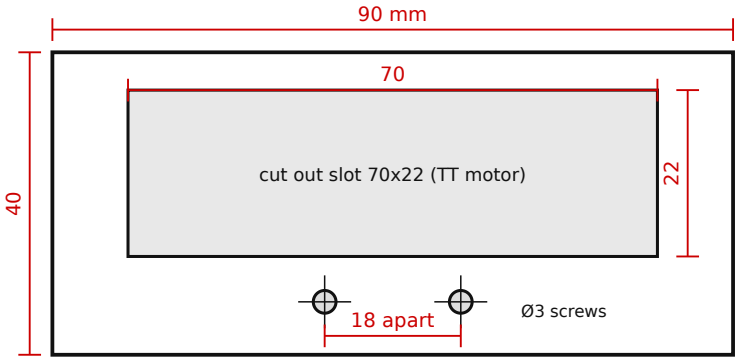


Crank disc 1 **Crank disc 2**

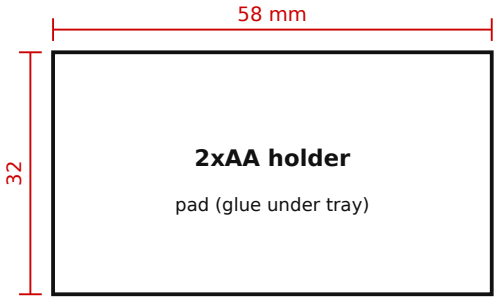
centre Ø4 (axle), Ø3 at 6 mm off-centre
lobe keeps the Ø3 hole inside the disc



Wheel spacer ring



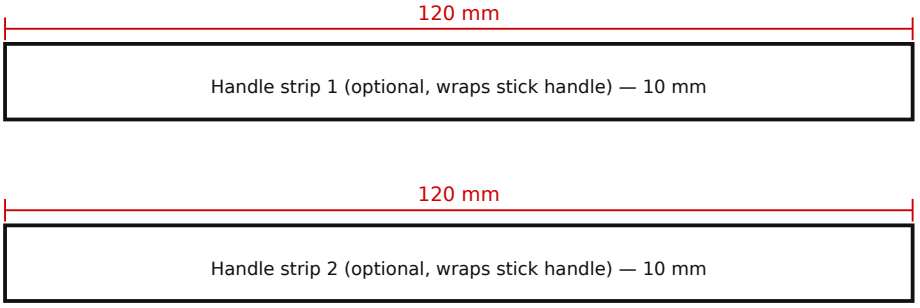
TT motor bracket



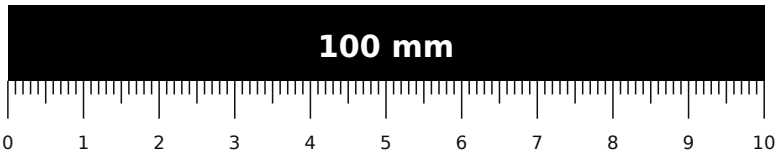
Battery pad 58x32



Hat brim



optional handle strips 120x10



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm.
If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.

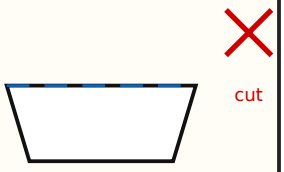
Robotics Project #13 · Templates 4/4 (1:1 scale)

Wheelbarrow Walker Robot

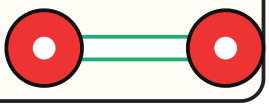
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — — — fold (dashed) ○ hole: grown-up pokes with a pencil/awl red = size in mm

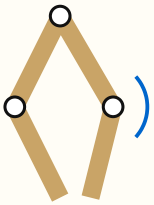
1 Cut and fold the barrow tray (sheet 1). Glue tabs inside.



2 Push skewer axle through a paper straw; glue a bottle-cap wheel on each end.



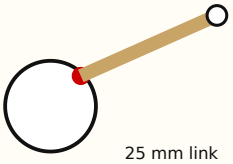
3 Pin thigh+shin and upper arm+forearm with Ø3 pins so they swing.



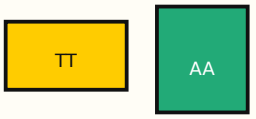
4 Glue the hands to the handles and the hip bar to the barrow frame.



5 Join the axle crank (off-centre Ø3 hole) to the knee with the 25 mm link.

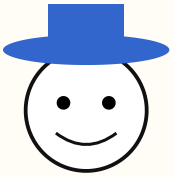


6 Add TT motor + 2xAA + slide switch — or skip it for push mode.

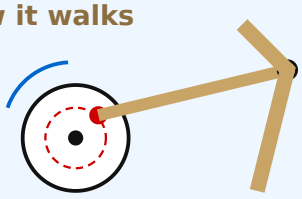


or push!

7 Add ping-pong ball head and bottle-cap hat. Test, then tune the link.



How it walks



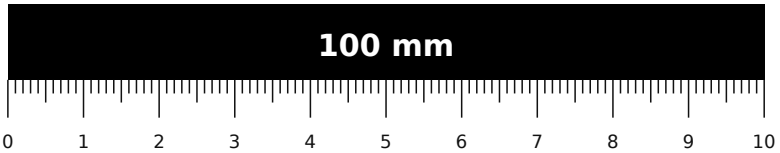
Wheel turns → off-centre pin goes round → link pushes knee back & forth → legs step!

Parts checklist

- [ ] 3 mm cardboard
- [ ] ~12 ice-cream sticks
- [ ] 2 bottle caps or 65 mm wheels
- [ ] Skewer / 3 mm rod
- [ ] Paper straw
- [ ] TT gear motor
- [ ] 2xAA holder
- [ ] Slide switch
- [ ] Ping-pong ball
- [ ] Bottle cap (hat)
- [ ] Hot glue (by a grown-up)

Safety first!

- A grown-up uses the hot glue gun.
- A grown-up does all craft-knife cutting and poking holes.
- Skewer tips are sharp: snip them.
- Batteries: put them in the right way round, never short the wires, take them out when finished.
- Keep small parts from little kids.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm. If not, reprint at 100% (Actual size). Use card ~1 mm (cereal box) for parts.

github.com/dev-repos/kids-robotics-projects · Kids.JoinDev · CC BY 4.0 · Grown-ups do all craft-knife cutting.